Université d'Ottawa
Faculté des sciences

University of Ottawa
Faculty of Science

PHY 3902: Physics and Applied Physics Laboratory I

Experiment 3: Properties of a laser diode

**** Work with a partner for this laboratory. You may submit one lab report per team.**

**** Your lab report should follow a standard format as described on Brightspace. Don't forget to address the various questions posed in this guide!**

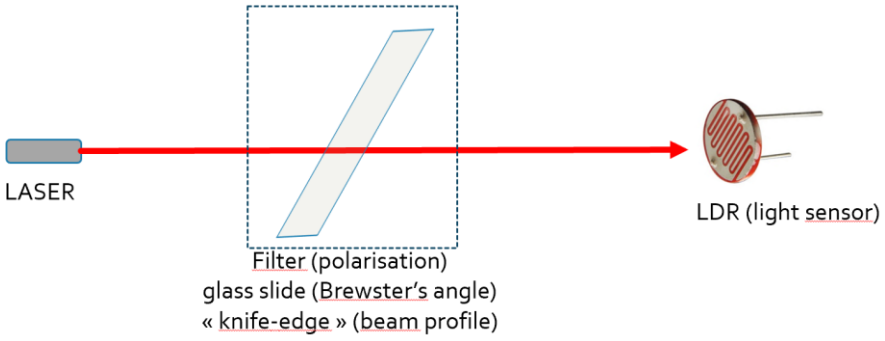
Objectives:

- Measuring the light intensity of the laser beam using a photoresistor
- Using the Arduino board to control a motor to manipulate the laser beam before measuring its intensity
- Putting together various components in a complete setup, and optimizing to obtain the best results
- Analyzing your results to characterize the properties of this laser diode

Caution: electronic devices not connected to ground are very sensitive to static electricity and can be rendered completely and irreversibly inoperative following a static discharge. Avoid working in dry carpeted rooms and discharge yourself regularly by touching a massive metallic object, copper water pipe etc.

Introduction

In this laboratory, you are going to characterize the properties of a laser diode by designing a set of experiments using the Arduino platform. The basic idea will be to measure the laser beam's intensity using a photodetector (i.e. photoresistor called LDR). However, the laser beam will be manipulated by inserting various filters and other equipment in between the laser diode and the photoresistor. The first experiment will allow you to explore the polarization of the laser beam. The second experiment will make it possible to quantify the Brewster's angle for glass. Finally, you will determine the beam's intensity profile.



The physics behind polarization and Brewster's angle are interesting and you should refresh your memories about these concepts. However, while your optics course might have presented these concepts in complicated mathematical terms, they are actually rather straightforward to observe in the lab. Consequently, the challenge here will not be "the physics", but rather to put together a setup that will allow you to quantify these phenomena. ***You will achieve this using the Arduino platform to measure the light intensity (MEASURE VOLTAGES) and to control motors (OUTPUT VOLTAGES) to move some filters. As you know, this type of process (control/measure) is a recurring theme in many data acquisition systems, not just when controlling a dinosaur!***

The instructions below are a guide rather than a complete set of instructions. When conducting scientific research, we do not always have complete guides telling us how to run experiments! I suggest you try to perform each experiment quickly and then refine your setup and your code in order to obtain the best results possible. Experimental research is an iterative process and it is up to you to figure out how best to run your experiments!

Methods

1. LASER

You will need to apply a voltage to the laser diode (650 nm) wires to turn it on! These laser diodes already have a series resistor integrated to the unit, so you can connect it directly to the voltage source. You have some choices: The easiest would be to connect the laser to the constant 5V output of the Arduino (the other wire will be connected to the Arduino's ground). However, the laser would be "ON" constantly, even when you are not making a measurement. An alternative approach might be to use one of the DIGITAL OUTPUT channels of your Arduino to apply 5V to the laser diode. This way, you can use the pinMode() and digitalWrite() commands in your Arduino code to turn "ON" the laser just before you start your measurements, and then turn it "OFF" when the measurements are done. Of course, if you want to control the laser intensity, you might be tempted to use your "DAC shield" (analog output voltage) in order to apply a desired DC voltage (between 0V and 5V) to light up the laser at a desired intensity. However, if you investigate how much current the DAC chip can supply, you will likely notice that it cannot supply enough power to light the laser diode. Please do not use the DAC chips to power the laser diode! You could use the PWM output of the Arduino to control the intensity of the laser if you feel this is necessary.

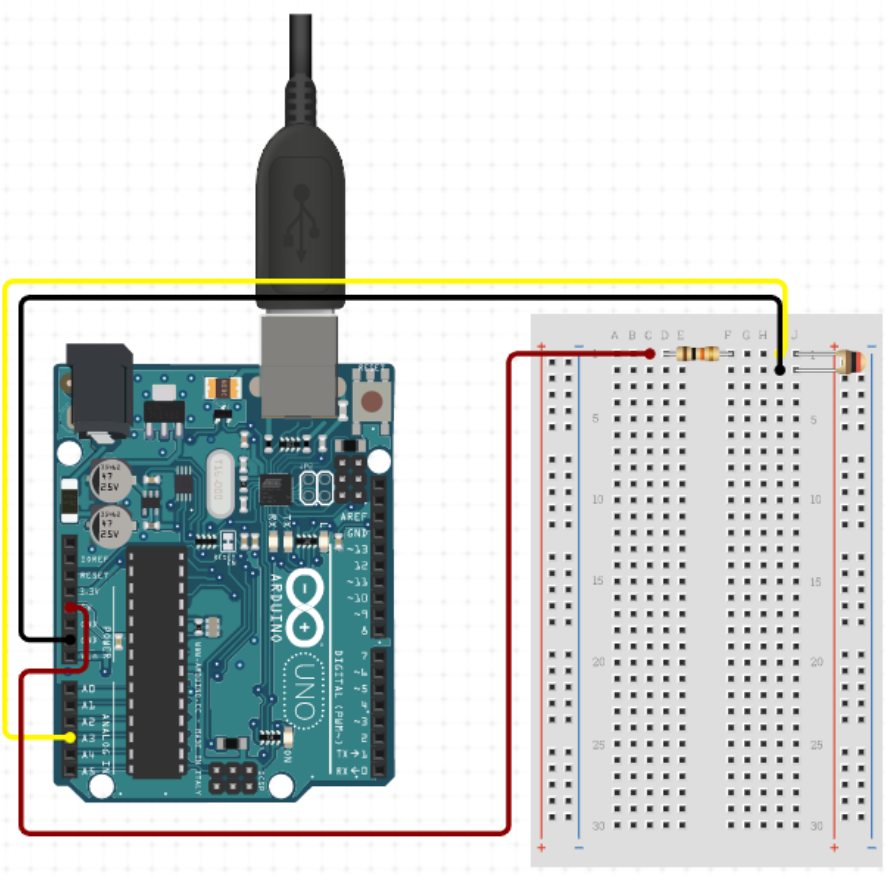
In any case, you might notice that the laser is too powerful and will saturate the photodetector. That is, if the laser intensity reaching the photodetector is too high, the photoresistor will reach its maximum value and become non-linear.

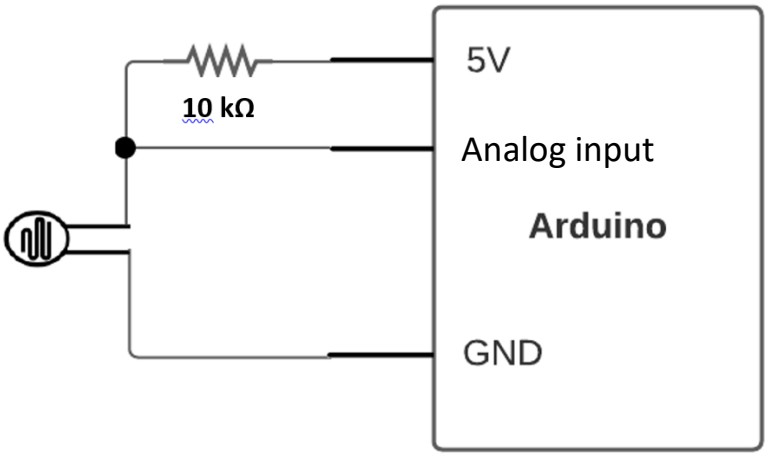
Note that the little laser diodes have a built-in lens, so you can rotate the lens to focus the beam as you wish. This will allow you to ensure that the entire laser beam is captured by the LDR detector. Pretty fancy for a $2 laser, right?!

2. PHOTORESISTOR (LDR)

The photoresistor is a little sensor whose resistance changes as a function of the light intensity impinging upon it. If you measure its resistance, you can infer the light intensity. You used this photoresistor (LDR) earlier, but here are some details to refresh your memory! We need to measure the resistance of the

photoresistor, since it is dependent on the light intensity. A simple way to measure the resistance is to set up a voltage divider (resistor and photoresistor in series). Use the 5V output of the Arduino to feed the circuit, and measure the voltage drop across the photoresistor. You should be able to relate the measured voltage to the detector's resistance, which in turn is proportional to the light intensity.





Voltage divider (with equation $R_{LDR} = \frac{V_0}{I} = V_0 * \frac{R}{(5V - V_0)}$)

Hints:

-Make sure that the laser beam is focused such that ALL the laser light is captured by the photoresistor. You can use the laser's built-in lens or an external lens to do this. Why is it important to do this?

-What's the best series resistance to use? Is 10 kΩ a good choice?

-Is it important to convert the measured voltage to a light intensity in proper units?

-It is possible that the laser diode intensity will be strong and thus saturate the photoresistor. How will you determine if you are saturating your photoresistor? How will you mitigate this issue?

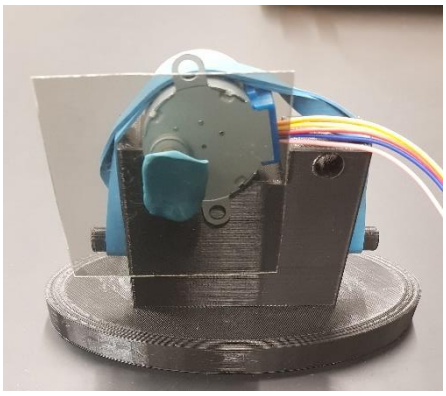
WARNING: Do not stare directly into the laser beam!

3. POLARIZATION

You have access to some polarizing filters. They look like basic grey filters, but the light transmitted through these filters will be linearly polarized. As you know, light is an electromagnetic wave. For polarized light, the electric field will be aligned in a single direction, perpendicular to the longitudinal direction. Of course, there is more theory than presented here, so feel free to explore the topic further on your own.

As a first step, determine if the beam emerging from your laser diode is polarized. An easy way to do this will be to aim your laser beam onto a piece of white paper. Then, make the beam pass through one of your polarizing filters and rotate the filter (by hand!) while observing the spot intensity on the piece of paper. If the spot intensity changes as you rotate the filter, then the laser diode beam is at least partially polarized! If the laser beam is not polarized, you can actually get a similar result by passing the laser beam through TWO polarizing filters. If you rotate one filter relative to the second, you should notice the transmitted intensity increase/decrease as a function of the relative angle between the two filters. When the two polarizing filters are aligned in the same direction, the transmitted intensity will be maximized. When the two filters are aligned perpendicularly to one another, the transmitted intensity will be minimized (zero ideally*). Note that adjusting the angle of a polarizing filter is a good way to control the intensity of the laser beam. You can use this technique to make sure you are not saturating the photoresistor signal!*

Now, you will use the Arduino to monitor the intensity of the transmitted polarized beam. Aim the laser beam towards the photoresistor. Make sure the beam is focused entirely onto the detector, using the integrated lens or an external lens. ***Add a polarizing filter in front of the laser diode. Mount a second polarizing filter onto the end of a stepper motor.***



Program your Arduino to rotate the filter by some angular increments, and then measure the intensity of transmitted light at each increment. Acquire data over a full rotation. The measured intensity should be a sinusoidal-like curve as a function of angle.

Hints:

-Make sure the laser beam is captured entirely by the detector throughout the experiment. Adjust the focus accordingly. It is possible that the laser beam deflects a little if your filter is not aligned properly.

-Figure out a way to calibrate (degrees per step) the rotation of the stepper motor. You might have done this already during the INTRO lab. In your Arduino code, you will tell the stepper motor to rotate by a certain number of "steps", but you should express your data as a function of degrees (or radians). Your report should highlight this calibration. Also, ask yourselves how many data points you want to acquire to obtain a nice, complete curve.

-In order to reduce noise and get better results, it is often useful to measure the signal many times (20-100 times) and then take an average of the intensity for each angle. You can also quantify the standard deviations and standard errors of your measurements. This was discussed in previous labs. Why would you need to do this here?

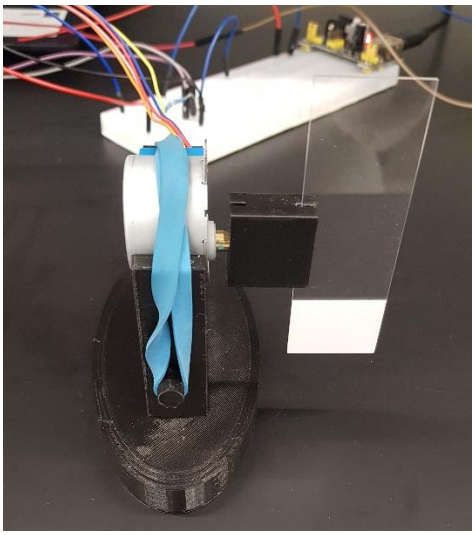
-Fit your data to an expected function. Use Python to do this as you are familiar with it now! Does it fit your data perfectly? If not, why?! Try to optimize this and discuss it in you report.

4. BREWSTER'S ANGLE

Brewster's angle is the angle of incidence at which light with a particular polarization is perfectly transmitted through a transparent dielectric surface, with no reflection! This effect is quite surprising as you might expect a small amount of reflected light, even for transparent materials such as glass. This effect is noticeable if you happen to have sunglasses with polarized lenses. When looking at the surface of water (a lake) during a sunny day, it might be very hard to see the fish in the water if the sun is out, but your polarized sunglasses allow you to see into the water since the reflected sunlight will be polarized, and your polarized sunglasses will block the remaining transmitted light before it reaches your eye.

Before using your Arduino to make this measurement, verify that this works "by hand". Aim the laser beam towards a glass slide (make sure the laser beam is already polarized!). Then, use a piece of white paper to see the reflected spot. Even if the glass is transparent, about 4% of the beam will be reflected. Change the angle of the glass slide and track the reflected spot. At the Brewster's angle (about 50-60 degrees), the reflected spot should disappear! If you do not see the reflected spot disappear, then it is likely that the laser beam is polarized in the wrong orientation. Try rotating the polarizing filter by 90 degrees and try again. If you still cannot notice this effect, then it is possible Sir David Brewster lied to us!

You can probably determine Brewster's angle by hand rather accurately. However, it is not as fun as doing it with the Arduino! Using a similar setup as above, replace the rotating polarizing filter with a rotating glass slide.

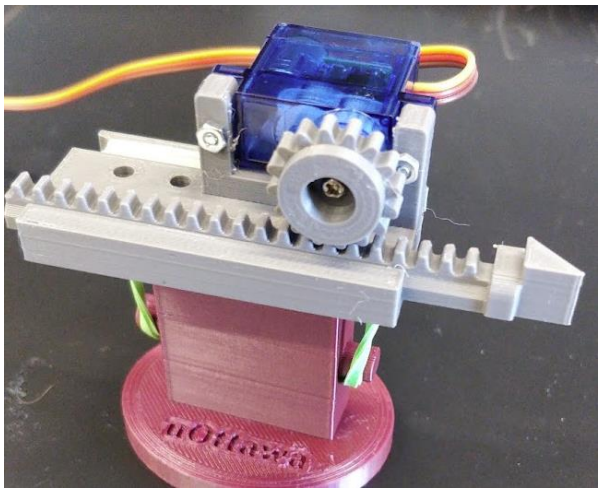


You can use the same stepper motor as before, so your Arduino code will not be much different. Of course, you will be measuring the intensity of the transmitted beam and not the reflected beam. That's ok since the transmitted beam intensity (T) will be maximum when the reflected beam intensity (R) is zero (R + T = 1). Therefore, you will still be able to determine the Brewster's angle for glass by measuring the transmitted light intensity.

5. LASER BEAM PROFILE

Semiconductor laser diodes usually have an oblong emitter shape. This is the main reason why the beam emitted by a laser diode is often oblong and not circular. You are likely to notice that your cheap laser diodes have rather messy beam shapes. Nevertheless, they should be oblong (ellipse)…. more or less! Let's use the Arduino to characterize the beam profile. You should be able to determine the ratio of the long axis and the short axis of the beam profile.

Use the same laser diode and the same photoresistor setup as done before. Now, we will determine the intensity distribution of the laser beam. To do this, mount a "straight edge" on a linear movement device, and monitor the laser beam intensity while the knife edge moves across the beam.



With proper calibration of the distance moved per step, the laser beam intensity distribution can be extracted.

Hints:

-What you actually measure is the integrated intensity distribution. Therefore, you will need to convert your data (take derivative) to reflect the actual intensity distribution. You can do this in Excel if you want, but the "cool kids" use Python! From these data, you can quantify the width of the beam (in micrometres). Do this for the long axis and the short axis.

-We 3D printed an adapter that will allow you to use your SERVO motor to induce a linear movement. The resolution and reproducibility of this linear actuator is not great. On the other hand, it was cheap (and fun!) to make! You should assess how well it works.

-Remember that the Arduino code (including library) will be different for the Servo compared to the stepper motor. Of course, you already know this!

-***You will certainly want to adjust the lens to increase the size of the laser beam, so that it is easier to measure its profile.***